

Agenda

- JDBC

JDBC

- RDBMS understand SQL language only.
- JDBC driver converts Java requests in database understandable form and database response in Java understandable form.
- JDBC drivers are of 4 types

1. Type I - Jdbc Odbc Bridge driver

- ODBC is standard of connecting to RDBMS (by Microsoft).
- Needs to create a DSN (data source name) from the control panel.
- From Java application JDBC Type I driver can communicate with that ODBC driver (DSN).
- The driver class: `sun.jdbc.odbc.JdbcOdbcDriver` -- built-in in Java.
- database url: `jdbc:odbc:dsn`
- Advantages:
- Can be easily connected to any database.
- Disadvantages:
- Slower execution (Multiple layers).
- The ODBC driver needs to be installed on the client machine.

2. Type II - Partial Java/Native driver

- Partially implemented in Java and partially in C/C++. Java code calls C/C++ methods via JNI.
- Different driver for different RDBMS. Example: Oracle OCI driver.
- Advantages:
- Faster execution
- Disadvantages:
- Partially in Java (not truly portable)
- Different driver for Different RDBMS

3. Type III - Middleware/Network driver

- Driver communicate with a middleware that in turn talks to RDBMS.
- Example: WebLogic RMI Driver
- Advantages:
- Client coding is easier (most task done by middleware)
- Disadvantages:
- Maintaining middleware is costlier
- Middleware specific to database

4. Type IV

- Database specific driver written completely in Java.
- Fully portable.
- Most commonly used.

- Example: Oracle thin driver, MySQL Connector/J, ...

MySQL Programming Steps

- step 0: Add JDBC driver into project/classpath. In Eclipse, project -> right click -> properties -> java build path -> libraries -> Add external jars -> select mysql driver jar.
- step 1: Load and register JDBC driver class. These drivers are auto-registered when loaded first time in JVM. This step is optional in Java SE applications from JDBC 4 spec.

```
Class.forName("com.mysql.cj.jdbc.Driver");
```

- step 2: Create JDBC connection using helper class DriverManager.

```
// db url = jdbc:dbname://db-server:port/database
Connection con =
DriverManager.getConnection("jdbc:mysql://localhost:3306/classwork", "root",
"manager");
```

- step 3: Create the PreparedStatement.

```
String sql = "SELECT * FROM students WHERE name=?";
PreparedStatement stmt = con.prepareStatement(sql);
```

- step 4: Execute the SQL query using the statement and process the result.

```
String sql = "non-select query";
int count = stmt.executeUpdate(sql); // returns number of rows affected
OR
String sql = "select query";
ResultSet rs = stmt.executeQuery(sql);
while(rs.next()) // fetch next row from db(return false when all rows completed)
{
    x = rs.getInt("col1");
    // get first column from the current row
    y = rs.getString("col2");
    // get second column from the current row
    z = rs.getDouble("col3");
    // get third column from the current row
    // process/print the result
}
rs.close();
```

- step 5: Close statement and connection.

```
con.close();  
stmt.close();
```

MySQL Driver Download

<https://mvnrepository.com/artifact/com.mysql/mysql-connector-j/8.1.0>

SUNBEAM